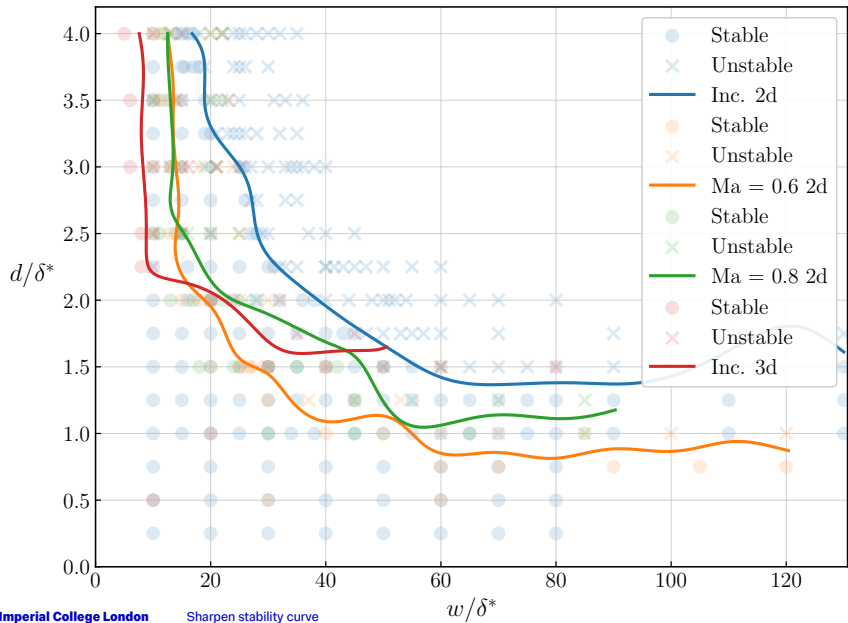


IMPERIAL

Sharpen stability curve

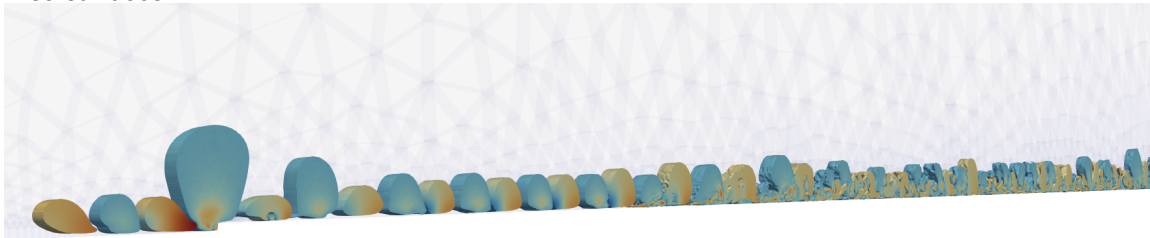
Víctor Ballester
December 18, 2025



3d TS transition ($d=1.5$ $w=80$)

- At shallow gaps the frequency of the modes at the downstream edge of the gap gets very close to the TS frequency regime. So it make sense.
- Turbulence not well resolved but the trend is clear.
- Mode in the cavity is 2d (the TS waves do not come from upstream).

v iso-surfaces



w contours



Burst of TS wave packets at shallow depths

- From time to time, a packet of TS waves is generated in the middle of the gap and travels downstream.
- Then for some time the flow stabilizes until a new packet is generated.
- This is for $d = 1.5$, $w = 50$ at $Ma = 0.6$.
- I have a video at [videos/d1.5_w50_comGapRe1000Ma0.6.gif](#).

